

1. [10 pts.] Evaluate $\lim_{x \rightarrow 0} \frac{3x}{\tan x + 2 \sin(5x)}$, if it exists.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{3x}{\tan x + 2 \sin(5x)} &= \lim_{x \rightarrow 0} \frac{1}{\frac{\tan x}{3x} + \frac{2 \sin(5x)}{3x}} \\ &= \lim_{x \rightarrow 0} \frac{1}{\frac{\tan x}{3x} + \frac{2 \sin(5x)}{3x}} \\ &= \frac{1}{\lim_{x \rightarrow 0} \frac{\tan x}{3x} + \lim_{x \rightarrow 0} \frac{2 \sin(5x)}{3x}} \\ &= \frac{1}{\frac{1}{3} + \frac{10}{3}} = \frac{3}{11} \end{aligned}$$

2. [10 pts.] Find $f'(0)$ for $f(x) = (\ln(x^2 + 1) + \cos(3x) + x)^3$.

We have $f'(x) = 3 (\ln(x^2 + 1) + \cos(3x) + x)^2 \left(\frac{2x}{x^2 + 1} - 3 \sin(3x) + 1 \right)$. Thus $f'(0) = 3(\ln 1 + \cos 0)^2 = 3$.

3. [5 + 5 + 5 = 15 pts.] Let $f(x) = \tan^{-1}(x + 1)$.

- (a) Find the linearization of f at $x = 0$.
- (b) Find the differential of f with $dx = 0.1$ at $x = 0$.
- (c) Use a linear approximation to estimate $\tan^{-1}(1.1)$.

We have $f(0) = \tan^{-1} 1 = \frac{\pi}{4}$ and $f'(x) = \frac{1}{1 + (x + 1)^2}$. Thus we obtain $f'(0) = \frac{1}{2}$.

(a) $L(x) = f(0) + f'(0)(x - 0) = \frac{\pi}{4} + \frac{1}{2}(x - 0) = \frac{\pi}{4} + \frac{x}{2}$.

(b) $dy = f'(0)dx = \left(\frac{1}{2}\right)(0.1) = 0.05$.

(c) $\tan^{-1}(1.1) \approx L(0.1) = \frac{\pi}{4} + \frac{0.1}{2} = \frac{\pi}{4} + 0.05$.

4. [15 pts.] Find an equation of the tangent line to the given curve at $x = 0$
 $y = \cosh(x^2 + x) + \ln(1 + xy)$.

It is easy to see that $y|_{x=0} = \cosh 0 + \ln 1 = 1 + 0 = 1$. Now we have $\frac{dy}{dx} = (2x+1) \sinh(x^2+x) + \frac{x \frac{dy}{dx} + y}{1+xy}$. Therefore, we obtain $\frac{dy}{dx}|_{(0,1)} = 1$. An equation of the tangent line is $y - 1 = (x - 0)$ or simply $y = x + 1$.

5. [10 pts.] The radius of a circle is increasing at a rate of 1 cm/min. At what rate the area of the circle is increasing when the area is π cm²?

We have $A = \pi r^2$ which implies that $r = 1$ if $A = \pi$. Therefore, we obtain $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$. This implies $\frac{dA}{dt}|_{r=1} = 2\pi$ cm²/min.

6. [10 pts.] Let $h(x)$ be a differentiable function satisfying $h(15) = 123$ and $h'(x) \leq 10$ for all real numbers x . What is the smallest possible value for $h(6)$?

We define the function h on the interval $[6, 15]$. Then it is easy to see that h is continuous on $[6, 15]$ and differentiable on $(6, 15)$ since h is always differentiable. Therefore by the Mean Value Theorem, there exists $c \in (6, 15)$ such that $h'(c) = \frac{h(15) - h(6)}{15 - 6} = \frac{123 - h(6)}{9}$. Now $h'(c) \leq 10$ which implies that $123 - h(6) \leq 90$ or simply $33 \leq h(6)$. Thus the smallest possible value of $h(6)$ is 33.

7. [4 + 6 + 7 + 7 + 6 = 30 pts.] Let $f(x) = \frac{x^2 - 9}{x^2 - 4}$.

(a) Find the x and y -intercepts of f .

The y -intercept is $y = f(0) = \frac{9}{4}$ and the x -intercepts are at $f(x) = 0$ or simply at $x = \pm 3$.

(b) Find the vertical and horizontal asymptotes for the graph of f , if any.

It is easy to see that $\lim_{x \rightarrow 2^\pm} \frac{x^2 - 9}{x^2 - 4} = \mp \infty$, $\lim_{x \rightarrow -2^\pm} \frac{x^2 - 9}{x^2 - 4} = \pm \infty$ and $\lim_{x \rightarrow \pm \infty} \frac{x^2 - 9}{x^2 - 4} = 1$. Thus we conclude that $x = \pm 2$ are vertical asymptotes and $y = 1$ is the only horizontal asymptote.

(c) Given that $f'(x) = \frac{10x}{(x^2 - 4)^2}$,

- i. find the intervals on which f is increasing and the intervals on which f is decreasing; It is clear that $f'(x) = 0$ or does not exist only at $x = 0$. Also, we see that $f'(x) > 0$ on $(0, 2)$ and on $(2, \infty)$ and $f'(x) < 0$ on $(-\infty, -2)$ and on $(-2, 0)$. Thus using the Increasing/ Decreasing Test f is increasing on $(0, 2)$, $(2, \infty)$ and decreasing on $(-\infty, -2)$, $(-2, 0)$.

- ii. find the local maximum and minimum values of f , if any.

Using the First Derivative Test, we conclude that f has local minimum value $f(0) = \frac{9}{4}$.

(d) Given that $f''(x) = -\frac{30x^2 + 40}{(x^2 - 4)^3}$.

- i. find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward;

It is clear that $f''(x) > 0$ on $(-2, 2)$ and $f''(x) < 0$ on $(-\infty, -2)$ and on $(2, \infty)$. Therefore using the Concavity Test, the function f is concave upward on $(-2, 2)$ and concave downward on $(-\infty, -2), (2, \infty)$.

- ii. find the points of inflection, if any.

There are no inflection points.

- (e) Sketch the graph of f .

